SERVICE 04 / INTEGRATION

Make the system *legible again.*

Trace hardware, firmware, dependencies and evidence until an incomplete prototype has a reviewable technical path forward.

REPOSITORY ANALYSIS   EVIDENCE RECOVERY   INTEGRATION
TECHNICAL DOCUMENTATION

01 / PROBLEMS THIS ADDRESSES

When context is *the missing part.*

Inherited systems often fail between layers: a build depends on undocumented setup, a communication path is unclear or a test result cannot be tied back to the code and hardware that produced it.

Typical starting points

Fragile dependencies, incomplete handoff, unclear failure reproduction, missing design decisions or a prototype that needs to become stable enough to review.

02 / TECHNICAL CAPABILITIES

Recover the links *between layers.*

The work starts with evidence: repositories, build instructions, interfaces, hardware access, logs and the people who know what the system was meant to do.

REPOSITORY

Code and dependencies

Build review, source organization, configuration context and reproduction steps around the current state.

SYSTEM

Interfaces and behavior

Hardware, firmware, communications, controls and tests connected into one technical explanation.

03 / ENGAGEMENT WORKFLOW

From unknowns *to a map.*

01

Set the boundary

Identify what is in scope and what evidence exists.

02

Inspect the system

Read the repository, hardware and dependencies.

Ø3

Reproduce the state

Confirm failures and observable behavior.

Ø4

Stabilize the path

Make targeted changes and tests reviewable.

Ø5

Document the handoff

Leave findings, limits and next actions clear.

PUBLIC TECHNICAL WORK

ESP32 Robotics Control Platform

A multi-layer system combining sensing, wireless input, PWM motor control, encoders, servo actuation and PID behavior.

VIEW PROJECT VIEW REPOSITORY ↗

PUBLIC CONTROLS WORK

TI C2000 Actuator Controller

F28069M command handling, state behavior, ePWM output, validation and fault response.

VIEW PROJECT   VIEW REPOSITORY ↗

PUBLIC RESEARCH WORK

TI Piccolo PMSM dynamometer

Dual-motor controls, CAN communications, real-time testing and validation evidence.

VIEW PROJECT   VIEW REPOSITORY ↗

PUBLIC POWER PROTOTYPE

5V / 1A SMPS

Offline flyback prototype with schematics, BOMs, waveforms and bench validation notes.

VIEW PROJECT   VIEW REPOSITORY ↗

04 / REPRESENTATIVE EVIDENCE

Public work with *clear boundaries.*

These projects are representative technical work, not completed client contracts. Each link points to the public repository rather than reproducing large source artifacts here.

05 / DELIVERABLES

Leave behind
a usable map.

- System and repository assessment
- Reproduction steps and root-cause findings
- Build, dependency or interface notes
- Stabilized prototype code and targeted tests
- Integration documentation and handoff notes

> **Recovery needs access.**
>
> The useful boundary depends on the hardware, code, documentation and people available to explain the current state.

START A TECHNICAL DISCUSSION

Bring the
unknowns.

Describe what is incomplete, what can be reproduced and which hardware, repositories or documentation are available. Secure file exchange can follow.

START A TECHNICAL DISCUSSION →

Or write directly: info@herdersystemen.com

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